

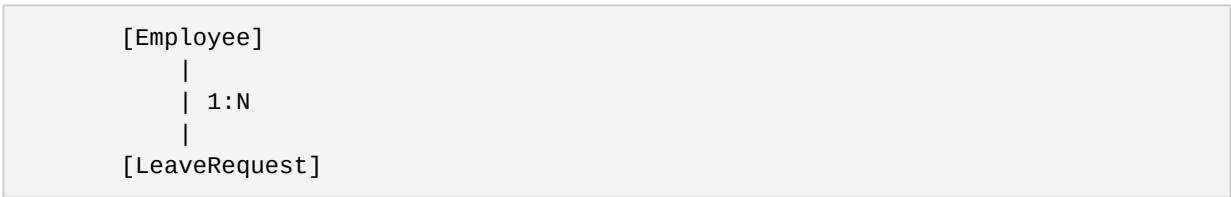
Data Model Documentation

Overview

The system manages the following core domains:

- Employee: Responsible for storing employee details and their leave requests.
- LeaveRequest: Responsible for managing leave requests submitted by employees.

Entity Relationship Diagram



Entities

Employee

Purpose: Represents an employee within the organization who can submit leave requests.

Table/Collection Name: Employee

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier for the employee
name	VARCHAR	Yes	Max 255 chars	Employee name
email	VARCHAR	Yes	Max 255 chars, unique	Employee email
created_at	TIMESTAMP	Yes	Auto-set on create	Record creation time
updated_at	TIMESTAMP	Yes	Auto-set on create/update	Last modification time

Relationships:

- Has many LeaveRequest (1:N): Inverse via LeaveRequest.employee_id

Validation Rules:

- name must be non-empty
- email must be non-empty and unique across all records

LeaveRequest

Purpose: Represents a leave request submitted by an employee.

Table/Collection Name: LeaveRequest

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier for the leave request
employee_id	UUID	Yes	Foreign key	Employee who requested leave
leave_type	VARCHAR	Yes	Max 255 chars	Type of leave requested
start_date	DATE	Yes	Must not be in the past	Start date of leave
end_date	DATE	Yes	Must be after start_date	End date of leave
status	VARCHAR	Yes	'PENDING', 'APPROVED', 'REJECTED', 'CANCELED'	Leave request status
created_at	TIMESTAMP	Yes	Auto-set on create	Record creation time
updated_at	TIMESTAMP	Yes	Auto-set on create/update	Last modification time

Relationships:

- Belongs to Employee (N:1): Foreign key employee_id references Employee.id

Validation Rules:

- employee_id must be a valid employee ID
- leave_type must be one of the predefined leave types
- start_date must be a valid date, must not be in the past

- end_date must be a valid date, must be after start_date
- status must be one of the predefined statuses

Relationships and Constraints

Foreign Keys

From Entity	Field	To Entity	Field	Cardinality	Delete Behavior
LeaveRequest	employee_id	Employee	id	N:1	CASCADE

Unique Constraints

- Employee(email): Ensures no duplicate email addresses

Data Validation Rules

Business Rules:

- leave_type must be one of the predefined leave types: 'SICK', 'VACATION', 'MATERNITY', 'PATERNITY'.
- start_date must be a valid date and must not be in the past.
- end_date must be a valid date and must be after start_date.

Format Rules:

- email: Must match the regex `^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$`

Data Lifecycle

Creation

- A new LeaveRequest is created when an employee submits a leave request.
- Default status is PENDING.
- created_by is automatically set to the current user.

Updates

- LeaveRequest.status can only be updated by an authorized user.
- updated_at is automatically updated on any modification.

Archival / Soft Delete

- Records are not archived but can be marked as CANCELED.

Indexes

- Primary key: id
- Foreign keys: employee_id
- Search fields: email

Example Records

Employee

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "name": "John Doe",
  "email": "john.doe@example.com",
  "created_at": "2026-01-15T10:30:00Z",
  "updated_at": "2026-01-15T10:30:00Z"
}
```

LeaveRequest

```
{
  "id": "123e4567-e89b-12d3-a456-426614174000",
  "employee_id": "550e8400-e29b-41d4-a716-446655440000",
  "leave_type": "SICK",
  "start_date": "2026-01-15",
  "end_date": "2026-01-20",
  "status": "PENDING",
  "created_at": "2026-01-15T10:30:00Z",
  "updated_at": "2026-01-15T10:30:00Z"
}
```